

Hidden Faults

Uncovering New Zealand's Seismic Story Through Data

Table of Content

Report Structure	3
Introduction	4
Glossary	5
Key Findings	6
1 - The Fault Nobody Knew About	6
2 - A Fault Holding Its Breath	7
3 - Frequency vs Energy – Two Different Stories	8
Discussion	9
Methodology	10
Data sources and Preprocessing	14
Spatial Grid Creation & Aggregation	15
Trend Analysis	16
Analytical Methods	19
Spatial Density Estimation (SDE)	19
Spatial Autocorrelation (Moran's I)	20
Limitations for Spatial Analysis	22
Machine Learning	23
What we tried to do	23
What is & Why RandomForest	23
How the model works	24
What we found	25
What this means	25
Further Improvements/Future Work	26
Highest Gain, Low Effort	26
Medium Gain, Medium Effort	27
Low Gain, High Effort	27
Acknowledgements	28

Report Structure

There will be two sections for this report:

Section 1 — General Audience

- Introduction — what, why, where
- Key findings — no jargon, visuals do the talking
- Discussion - What it means for NZ

Section 2 — Technical Detail

- Methodology
- Data Sources and Preprocessing
- Analytical Methods
- Statistical Outputs
- Limitations and future work

This report is structured to address two different audiences. The first section is written in a clear and accessible way, allowing general readers to understand the key findings without requiring technical background knowledge. In parallel, more detailed technical parts are described in the second section for readers who want a deeper understanding of the methods and analysis.

Introduction

New Zealand sits on one of the most geologically active regions on Earth. Positioned at the boundary of the Pacific and Australian tectonic plates, the country experiences thousands of earthquakes every year — most too small to feel, but some capable of reshaping entire cities overnight.

This project analyses 25 years of earthquake data across New Zealand, from 2001 to 2026, collected over 74000 earthquakes, covering every recorded event of magnitude 3.0 or greater. Using spatial analysis and geographic information systems (GIS), it examines not just where earthquakes happen, but how frequently, how intensely, and whether activity is changing over time.

The goal is not prediction — earthquake prediction remains one of science's greatest unsolved problems. Instead, this project asks a simpler but equally important question: **what does a quarter century of seismic data actually tell us about where New Zealand shakes, and why?**

Glossary

The following terms are used throughout this report. Definitions are provided here for readers who may be unfamiliar with the technical language used in spatial and seismic analysis.

Fault Line

A crack in the Earth's crust where two sections of rock have shifted against each other. When stress builds up along these cracks and suddenly releases, the result is an earthquake.

Spatial Density Estimation Map (SDE)

In plain terms, it is a technique that divides the study area into a uniform grid of cells and counts how many earthquakes fall into each cell, then applies smoothing to blend neighbouring cells together into a continuous surface. Rather than showing each earthquake as a single dot, it reveals the overall pattern — the brighter the colour, the more seismic activity in that area.

Magnitude

Magnitude is a measure of the size of strength of an earthquake. It is recorded on a logarithmic scale, meaning each whole number increase represents a much larger earthquake, not just a small step up.

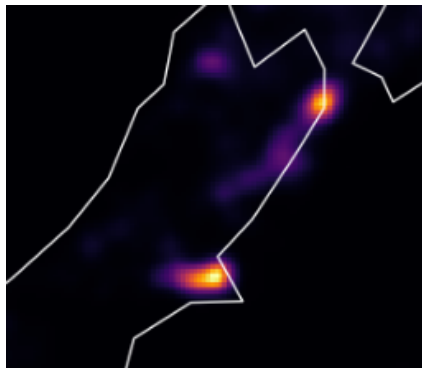
It is calculated on a logarithmic scale, meaning each increase of one unit represents about 10 times greater ground motion and roughly 30 times more energy release. This means a magnitude 7 earthquake is not just slightly stronger than a magnitude 6—it is dramatically more powerful.

Key Findings

1 - The Fault Nobody Knew About

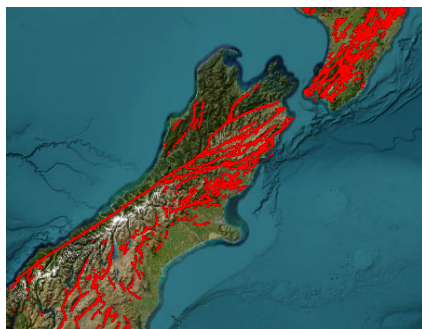
During the process of verifying fault lines, a bright spot was identified over Christchurch city. When it was compared to GNS Science's New Zealand Active Faults Database (NZAFD), there wasn't one shown beneath Christchurch.

The heatmap was initially thought to have been incorrectly generated, but it was later found out that the fault was completely hidden underground with no surface evidence of its existence – hence it didn't appear on the GNS map. It only became known to geologists when it ruptured in 2010-2011. Yet the 25 years of earthquake data in this project independently reveals its location as a clear hotspot.



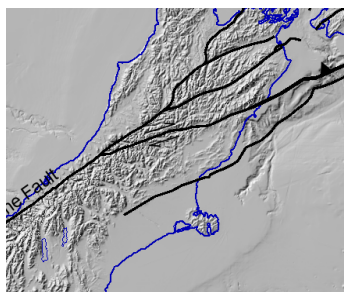
“Here's what was found”

Dark-Background Heat Map: SDE Earthquake Density Surface that was generated in Jupyter Notebook. Shows a bright spot beneath Christchurch city.



“Here's what the official map shows”

Red Active Faultline Map: GNS Science New Zealand Active Faults Database (NZAFD) (data.gns.cri.nz/af) showed no active fault running under Christchurch



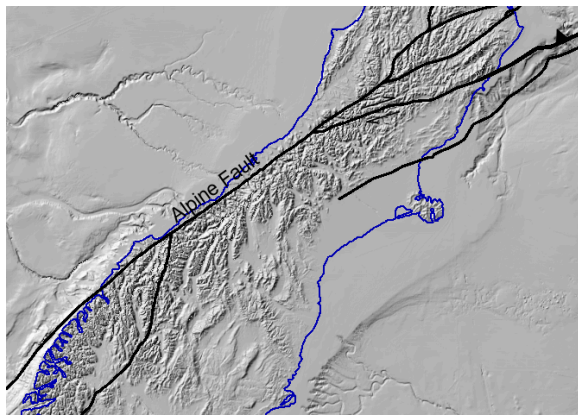
“Here's the cross-reference confirming it”

Grey-Blue-Black Faultline Map: Cross-referencing (data.gns.cri.nz/tez) against the NZAFD map confirmed no recorded fault line at this location.

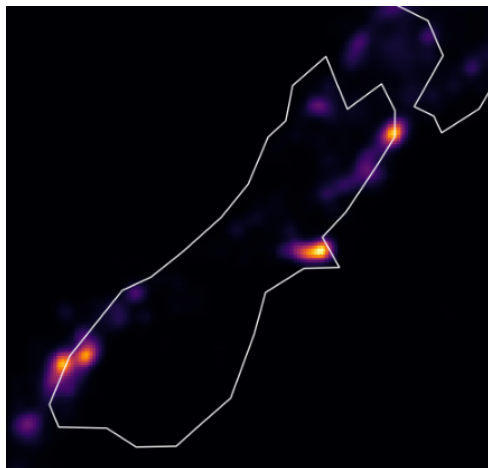
The GNS Active Faults Database is current and maintained - it does not show a fault beneath Christchurch because the database deliberately excludes blind faults with no surface expression. The 2010-2011 Canterbury earthquake sequence revealed a fault that, by definition, could not have appeared on any geological map beforehand.

This project's earthquake density analysis independently identified the same location as a clear hotspot – detecting through seismic data what surface geology cannot see.

2 - A Fault Holding Its Breath



The Alpine Fault is a well-known major geological feature in New Zealand. It runs almost the entire length of the South Island and is one of the most dangerous faults in NZ – capable of a magnitude 8+ earthquake. However, on the trend map and SDE that was produced, it barely shows up. There was barely any seismic activity going on.



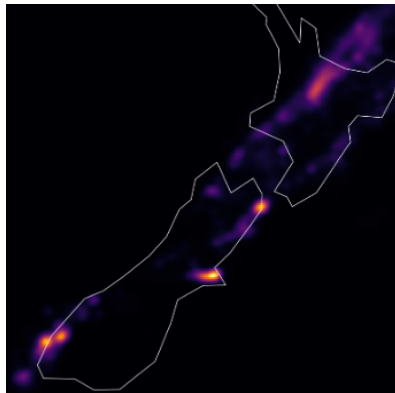
< SDE Image

There are no bright cells along South Island's West Coastline, where the Alpine Fault runs through. It shows a relatively low density along the fault compared to Christchurch and Fiordland.

“That silence isn’t reassuring. It means the fault is locked – two tectonic plates grinding against each other with nowhere to go, building up pressure year by year. The last major rupture was in 1717. Scientists estimate there’s a 75% chance of another magnitude 8+ before 2068.” ([GNS Science AF8 programme](#)).

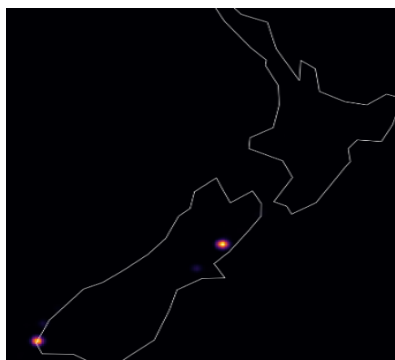
3 - Frequency vs Energy – Two Different Stories

Imagine counting traffic. You could count the number of vehicles, or measure the total weight of all vehicles. However a busy street with thousands of bicycles can look different from a quiet highway with a few freight trucks. Neither count is wrong – they just answer different questions.



A limitation of a frequency-based Spatial Density Estimation (SDE) map is that it does not account for earthquake magnitude. Instead, it represents only the spatial concentration of events—areas with brighter intensities indicate a higher number of recorded earthquakes, regardless of their individual strength.

1.



In contrast, a magnitude-weighted SDE incorporates the relative size of each event, typically using measures such as moment magnitude (M_w). This approach emphasizes regions where seismic energy release is greatest rather than where events are most frequent. As a result, the map becomes dominated by a small number of high magnitude earthquakes.

In the New Zealand context, this effect is clearly visible: major events such as the 2009 Fiordland earthquake (Dusky Sound, M_w 7.8) and the 2010-2011 Canterbury earthquake sequence strongly influence the output. Because these events release disproportionately more energy than smaller earthquakes, the weighted SDE produces only a few highly concentrated high-intensity regions, while areas with frequent but low-magnitude seismicity appear less prominent.

Discussion

So, what do all these findings mean? The results from this analysis highlight how earthquake patterns in New Zealand are not always obvious at first glance, and why different ways of visualising the data can lead to very different interpretations.

One key finding is the concentration of earthquakes beneath Christchurch. The maps show a clear cluster of seismic activity in this area, despite the absence of a clearly mapped surface fault line. This aligns with what is known from the Canterbury earthquake sequence (2010-2011), where several earthquakes were caused by previously unmapped or “blind” faults buried underground. These faults do not reach the surface, which makes them difficult to identify before they become active. This finding reinforces an important point: the absence of a visible fault line does not necessarily mean an area is low risk.

Another notable result is the relatively low level of earthquake activity along the Alpine Fault on the west coast of the South Island. In the frequency-based map, this region appears comparatively quiet, with fewer recorded earthquakes than other parts of the country. However, this does not mean the fault is inactive. The Alpine Fault is known to produce large earthquakes infrequently, meaning long periods of low activity can occur between major events. As a result, a quiet appearance on a short-term dataset (2001-2026) may reflect a gap between large earthquakes rather than a lack of underlying tectonic stress.

The comparison between frequency-based and magnitude-weighted maps further highlights how interpretation depends on what is being measured. The frequency map emphasises areas where earthquakes happen often, even if they are small. In contrast, the weighted map highlights where the largest earthquakes release the most energy. This causes a few major events—such as those in Fiordland and Canterbury—to dominate the visualisation, while regions with many smaller earthquakes become less prominent. Together, these two approaches provide complementary perspectives: one shows how often earthquakes occur, while the other shows where the most significant seismic energy is released.

Overall, these findings demonstrate that earthquake risk cannot be understood through a single map or metric. Areas that appear quiet may still be capable of producing major earthquakes, and regions without visible fault lines may still experience significant seismic activity. Using multiple analytical approaches provides a more complete picture of New Zealand’s seismic behaviour, while also highlighting the limitations of relying on any single method.

Methodology

Technical Workflow Overview

The analytical workflow for this project follows a structured geospatial data pipeline, progressing from raw data acquisition to final outputs:

1. Data Acquisition and Preparation

- Earthquake data was obtained from the GeoNet catalogue (quakesearch.geonet.org.nz), covering 2001–2026, magnitude ≥ 3 .
- Data was loaded with pandas and verified for completeness (date range, column integrity).
- A duplicate check was performed; no duplicate records were found, indicating the GeoNet export was already de-duplicated at the source.
- Date range was confirmed to match the intended study window (2001–2026).

2. Geospatial Conversion and CRS Validation

- Raw tabular data (lat/lon columns) was converted into a GeoDataFrame using GeoPandas, with point geometries constructed from coordinate columns.
- Coordinate Reference System (CRS) was set and, where necessary, reprojected to EPSG:4326 for storage.
- CRS was set to EPSG:2193 for accurate distance/area calculations in NZ.
- A data integrity validation pass was run on the resulting GeoDataFrame to check for duplicate earthquakes.

3. Spatial Grid Construction

- A uniform spatial grid (tessellation) was generated to divide the study area into equally sized cells of 5km x 5km.
- Grid polygons were generated and visually spot-checked (e.g. comparing cell 0 and cell 1 directly) to confirm the grid was constructed correctly — no degenerate (zero-area) cells, consistent CRS, no overlap or gaps.
- A quick visualization confirmed the grid aligned correctly with the NZ mainland extent.

4. Spatial Join — Assigning Earthquakes to Grid Cells

- Each grid cell was assigned a unique ID.
- A spatial join was performed to assign each earthquake event to its containing grid cell.
- The join was validated by checking both lengths of the geodataframe and joined results (74007 to 74007, lengths match - no events were duplicated or dropped by the join itself).
- Earthquake counts per grid cell were computed and attached back to the grid as an attribute.

5. Density Sanity Check (Pre-SDE)

- A first-pass density visualization was produced using log-scale earthquake counts per cell, to confirm known seismic hotspots were visible in the data before investing in formal SDE.

6. Temporal Slicing (Exploratory)

- Years were extracted from event timestamps.
- Earthquake counts per cell, per year were computed and merged back into the grid.
- A single year was visualized as a representative sample to sanity-check the per-year aggregation.
- Total earthquakes per year were plotted across the full study period.
- A 3-year rolling average was applied to the yearly totals to smooth short-term noise and reveal longer-term trend direction.

7. Spatial Density Estimation (First Formal ESDA)

- A 2D histogram heatmap was generated at 5km resolution, counting earthquake events per cell across the NZ mainland.
- Gaussian smoothing was applied post-binning (sigma=2, equivalent to ~10km smoothing radius) to produce a continuous-looking density surface.
- This produced the frequency-based (unweighted) density surface.

8. Moran's I (Second Formal ESDA)

- A spatial weights matrix was constructed using Queen contiguity (cells sharing any edge or corner treated as neighbors), applied only to cells with at least one earthquake event. Weights were row-standardised to account for edge cells having fewer neighbors.
- Global Moran's I was calculated on the per-cell earthquake counts.

9. Magnitude Weighting

- Each earthquake was assigned a weight proportional to its seismic moment, calculated as $10^{(1.5 \times \text{magnitude})}$, reflecting actual energy release rather than event count.
- The same 2D histogram with Gaussian smoothing approach ($\sigma=2$, 5km resolution) was applied with weights, producing a magnitude-weighted density surface.
- Results were compared side-by-side against the unweighted surface from Section 7.

10. Trend Analysis (Formal)

- Using the yearly per-cell counts established in Section 6, a linear trend (slope) was calculated for each grid cell over the study period.
- Trend values were merged back into the grid for spatial visualization.

11. Temporal Animation

- A single representative frame was built and tested before generating the full animation, to validate rendering correctness.
- A GIF animation was built and exported, visualizing how spatial earthquake patterns evolved over the 25-year study period.

12. Data Consolidation for Modelling

- Cleaned earthquake event data, the grid with aggregated earthquake counts, trend analysis results, and summary statistics were consolidated into a single feature set in preparation for predictive modelling.

13. Random Forest Classification

- Earthquake events were classified into three magnitude bins: M3-4 (62,491 events), M4-5 (10,454 events), and M5+ (1,062 events). M5-6 and M6+ were initially separate classes but merged due to insufficient sample size in the rarer categories.
- Features used: latitude, longitude, and depth — raw event properties only, not derived spatial features from earlier pipeline stages.
- Data was split 80/20 into training (59,205) and test (14,802) sets using a random split. Note: this does not account for spatial autocorrelation between nearby events, which may inflate apparent performance.
- A Random Forest classifier was trained (100 trees, balanced class weights to partially compensate for class imbalance, all CPU cores used for parallel training).
- Model performance: overall accuracy 0.85, macro F1 0.44. Per-class F1: M3-4 (0.92), M4-5 (0.34), M5+ (0.05). The high overall accuracy is misleading — it reflects dominance of the M3-4 class. The model performs poorly on M5+ events, the class most relevant to disaster risk.
- Feature importance was extracted and plotted.

Data sources and Preprocessing

Earthquake data for this project was obtained from the GeoNet catalogue, covering the period 2001-2026. The dataset includes over 74,000 recorded events of magnitude 3.0 or greater. The raw data was divided across four CSV files (2001-2004, 2004-2010, 2010-2016, 2016-2026), each containing fields such as event ID, time, location, magnitude, depth, evaluation method, and phase/station counts.

The preprocessing workflow consisted of the following steps:

1. Data Loading and Consolidation

- The four CSV files were imported using Python's *pandas* library and concatenated into a single dataset.
- Columns of interest were retained including *publicid*, *eventtype*, *origintime*, *longitude*, *latitude*, *magnitude*, and *depth*.

2. Data Cleaning

- Duplicate records were removed to ensure each earthquake is represented only once.
- The date range was confirmed to cover the full 2001-2026 period.
- The cleaned dataset was saved as a single GeoPackage for downstream analysis

3. Spatial Preparation

- Using *geopandas*, a geometry column was created from longitude and latitude coordinates, forming a GeoDataFrame
- The coordinate reference system (CRS) was initially set to EPSG:4326 (WGS84) and then reprojected to EPSG:2193 (New Zealand Transverse Mercator) for accurate spatial analysis.

4. Data Validation

- Duplicate earthquake events were checked via the *publicid* field, confirming zero duplicates.
- Basic integrity checks ensured all coordinates, magnitudes, and dates were consistent and within expected ranges.

This preprocessing pipeline ensured that the dataset was clean, consistent and spatially ready for the subsequent analytical steps.

Spatial Grid Creation & Aggregation

To analyse the spatial distribution of earthquakes across New Zealand, a uniform spatial grid was created to divide the country into manageable units for aggregation and analysis.

1. Grid Creation

- New Zealand was divided into 5 km x 5 km cells using *shapely.geometry* to generate polygon geometries.
- Grid bounds were determined based on the extent of the cleaned earthquake dataset, and a nested loop was used to perform 2D tiling over the study area.
- The resulting polygons were converted into a GeoDataFrame (`grid_gdf`) for further analysis.

2. Grid Identification

- Each grid cell was assigned a unique identifier to enable referencing and aggregation.

3. Spatial Join and Aggregation

- Earthquake events were assigned to their corresponding grid cells via a spatial join using *geopandas*.
- The join was validated by comparing the number of earthquakes before and after assignment, ensuring no data was lost.
- Aggregation was performed to count the number of earthquakes in each grid cell, producing summary statistics.

4. Summary Statistics:

Count - 126,492 / Mean - 0.585 / Std - 3.684 / Min - 0 / 25% - 0 / 50% - 0
75% - 0 / Max - 351

These results indicate that most grid cells experienced few or no earthquakes, while a small number of cells captured high concentrations of events, highlighting areas of significant seismic activity. This gridded dataset forms the foundation for subsequent Spatial Density Estimation, magnitude-weighting, and spatial autocorrelation analyses.

Trend Analysis

To examine how earthquake activity changes over time, a temporal trend analysis was conducted using yearly aggregations of the GeoNet dataset (2001-2026).

Think of it as “analytical preparation”, the output of this section is what feeds into the SDE, magnitude-weighted maps, Moran’s I, and animations.

Prior to formal trend analysis (Step 6), yearly earthquake counts were visualised across the study period including a 3-year rolling average, confirming sufficient temporal variation to warrant per-cell slope estimation.

1. Temporal Data Preparation

- The `origintime` field was converted into a datetime format to enable time-based operations.
- The year of each earthquake event was extracted and stored as a new attribute.

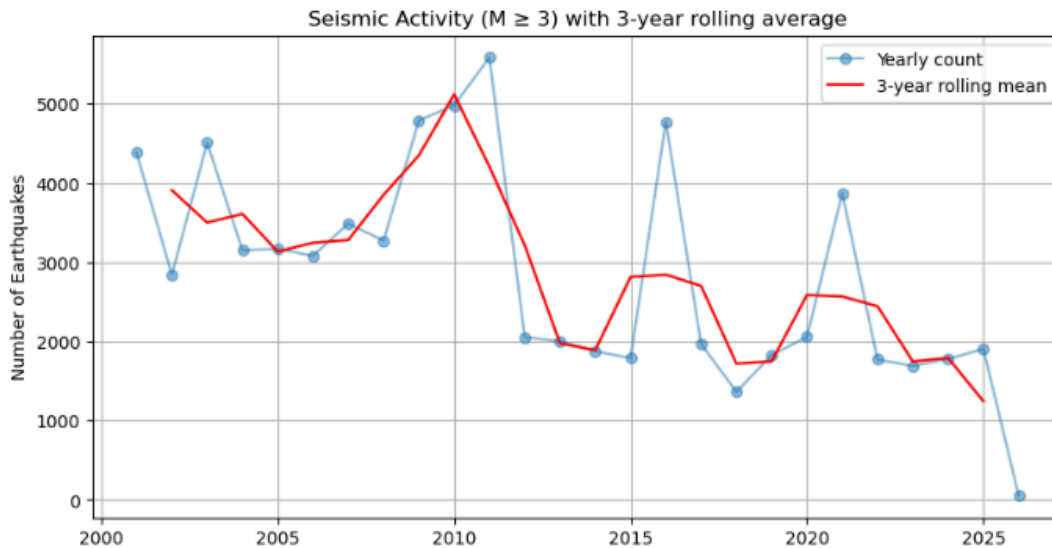
2. Yearly Aggregation

- Earthquake events were grouped by year to calculate the total number of earthquakes occurring annually.
- In addition, spatial aggregation was performed by counting the number of earthquakes per grid cell for each year.

3. Spatial-Temporal Integration

- Yearly earthquake counts were joined back to the spatial grid, producing a dataset that captures both location and time.
- This enabled the analysis of how seismic activity evolves spatially across different years.

4. Trend Visualisation



A time series graph was generated to display total earthquake counts per year, allowing for identification of peaks and fluctuations in seismic activity.

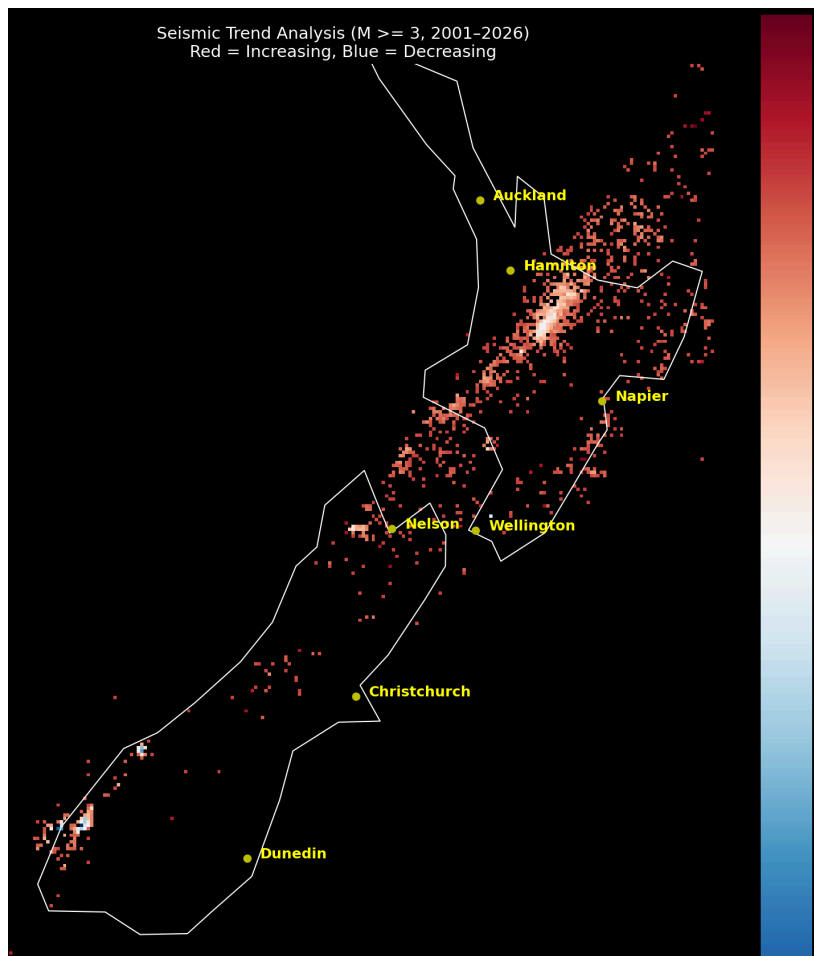
These patterns were further explored through temporal animation, showing how earthquake distributions shift over time across New Zealand.

1. Seismic activity is inherently volatile

- Earthquakes occur in bursts, swarms, or clusters along faults
- Year-to-year counts fluctuate a lot, even if long-term trends are flat.
- A single big quake (or swarm) can spike a year's total.

2. Long-term trend

- The overall trend indicates a slight decrease in total $M \geq 3$ events over the period.
- But with only ~25 years, a short-term volatility can mask real trends
- Statistically, 25 years is still a small sample for tectonic-scale analysis



Mean = -0.006 — on average, cells are very slightly declining in activity. Barely noticeable, almost flat.

Median = -0.003 — same story, most of NZ is essentially flat or marginally quieter.

Min = -1.98 — one cell is losing almost 2 earthquakes per year. Almost certainly a Canterbury aftershock cell — the 2010-2011 spike was so dramatic that the long decay afterward pulls the slope steeply negative.

Max = +0.43 — one cell is gaining 0.43 earthquakes per year. That's the most interesting number — somewhere in NZ a fault zone is genuinely becoming more active over 25 years.

The trend analysis heatmap highlights cells with increasing or decreasing earthquake activity over the 25-year period. While many cells appear red, indicating slight increases, the overall statistics show a nearly flat nationwide pattern.

Analytical Methods

Spatial Density Estimation (SDE)

was used to visualise the spatial intensity of earthquakes across New Zealand.

True KDE was attempted twice — first with `scipy.stats.gaussian_kde`, then with `sklearn.neighbors.KernelDensity` — both were too slow on 74,007 points. A 2D histogram with Gaussian smoothing was used as a faster approximation, producing a visually and analytically comparable result.

The steps were as follows:

1. Coordinate Extraction and Study Area Definition

- Earthquake coordinates were extracted from the cleaned `GeoDataFrame`.
- Mainland New Zealand bounds were manually defined in NZTM2000 coordinates (metres) to constrain the analysis.

2. Density Estimation

- A fast density estimation algorithm was applied to calculate earthquake intensity over a 2D grid.
- Gaussian smoothing was applied using `scipy.ndimage` to produce a continuous heatmap, reducing pixelation and highlighting clusters of events.

3. Reprojection and Visualisation

- The resulting density raster was reprojected to Web Mercator (EPSG:3857) for compatibility with basemaps in visualisation.
- *Numpy*, *geopandas*, and *matplotlib* were used to generate static maps, showing both frequency-based and magnitude-weighted densities.

4. Magnitude Weighting

- SDE was performed in two ways:
 - Frequency-based: all earthquakes contribute equally
 - Magnitude-weighted: each earthquake's contribution is scaled by its magnitude to highlight areas with high energy release

- Log scaling was applied to reduce visual distortion caused by extreme values, and zero-count grid cells were masked to avoid misleading representations.

Frequency-based SDE highlights regions of high earthquake occurrence but underweights the significance of rare, large events. Magnitude-weighted SDE accounts for energy release but risks obscuring zones of frequent low-magnitude activity. Both maps are therefore presented together to provide a complete picture of seismic hazards across the study area.

Spatial Autocorrelation (Moran's I)

was used to quantify the degree of spatial clustering of earthquakes across New Zealand.

1. Spatial Weights Matrix

- A spatial weights matrix was created using *libpysal*, based on adjacency between the 5km x 5km grid cells.
- Queen contiguity was applied, meaning that cells sharing edges or corners were considered neighbours.

2. Calculation of Moran's I

- The *esda* library was used to calculate Moran's I for both raw earthquake counts and magnitude-weighted counts.
- This statistic measures whether high or low values are clustered spatially, with positive values indicating clustering and negative values suggesting dispersion.

3. Interpretation

The results provide a numerical measure of clustering in earthquake occurrences, complementing the visual patterns observed in the SDE heatmaps:

Moran's I = 0.6695

On a scale of 0-1, this is very high. The earthquake hotspots are strongly clustered – active cells tend to be surrounded by other active cells. This is exactly what is expected from fault zones, which are long continuous structures, not random scatter.

P-value = 0.001

This means there is only a 0.1% chance this clustering is random. $p < 0.05$ is needed to be confident, but this result is at $p < 0.001$ – that's 50 times more confident than the minimum threshold. The clustering is statistically real beyond any reasonable doubt.

Z-score = 141.06

This is how many standard deviations the results sit away from what random scatter would produce. Anything above 3 is considered significant. This result is an almost impossibly strong signal. For context, finding a z-score this high basically means the clustering isn't just real, it's the dominant spatial feature of the entire dataset.

Limitations for Spatial Analysis

While this project provides a detailed a spatial and temporal analysis of New Zealand earthquakes over 25 years, several limitations should be acknowledged:

1. Data Limitations

- Only earthquakes with magnitude ≥ 3.0 were included, which excludes smaller events that may contribute to localised stress patterns.
- Data quality depends on the GeoNet network; older events or those in remote areas may have higher uncertainty in location, depth, or magnitude

2. Spatial Resolution

- The $5\text{km} \times 5\text{km}$ cell size was selected as an initial working resolution based on NZ's geographic scale, but was not formally derived from event density analysis. Sensitivity testing across different cell sizes was not conducted, meaning results may vary under alternative grid configurations.
- Some localised trends may be masked, particularly near small faults or urban areas with sparse seismicity.

3. Temporal Analysis Constraints

- Linear trend analysis assumes constant change per year, which may not fully capture episodic sequences like aftershock swarms.
- Short-term fluctuations can skew slopes in individual cells (e.g., Canterbury aftershock decay)

Machine Learning

What we tried to do

The question we wanted to answer was simple – given where an earthquake happens and how deep it is, can we predict how strong it tends to be?

Knowing that location predicts magnitude class better than depth tells us about how NZ fault systems work – different fault regimes produce characteristically different magnitude profiles. That's useful for long term hazard zoning, building codes, and infrastructure planning – even if it can't tell us when the next quake hits.

What is & Why RandomForest

If we tried to create a single decision tree that looks at the data and learns a series of yes/no rules:

“Is longitude > 174? If yes, is depth < 40km? If yes, predict M3-4.”

It's like a flowchart of questions that eventually lands on an answer.

The problem with one tree is it's fragile – slightly different training data processes a completely different flowchart.

Random Forest fixes this by building 100 different trees, each trained on a random subset of the data and features. Then they all vote.

It's like asking 100 different geologists to each independently study a random portion of the earthquake records and make a prediction.

Some will be wrong, but the majority vote of 100 independent experts is much more reliable than any single one.

“Random” refers to the random subsets each tree sees. “Forest” is just many trees together.

How the model works

The Model needed two things: features (the columns you feed in as inputs) and a target (the one column you're trying to predict). The features that were used were the latitude, longitude and depth of each earthquake, and the goal was predicting the magnitude class.

Steps we took to setup the model:

1. Defining magnitude bins

- Classify the earthquakes into several magnitude classes (3, 4, 5, 6 - 10).
- The reason all of M6+ was collapsed into one class rather than splitting it further (M6-M7, M7-M8 etc.) is that those events are so rare in the dataset that the model would have almost no examples to learn from.
- It was better to have a decent sample size than to split them and have almost nothing.

2. Machine Learning Preparations

- Added sklearn tools
- RandomForestClassifier (The actual model. This is what learns pattern from the data)
- train_test_split (A utility that splits data into a training set and a test set)
- classification_report (A utility that prints out how well the model performed)
- Defined features (latitude, longitude, depth) and target (mag_class)

3. Splitting the data into training and test sets

- 80% of the data trains the model, 20% is used to evaluate it at the end
- Training Set : 59205 events
- Test set: 14802 events

4. Train the Model

- n_estimators = 100 (the RF builds 100 individual decision trees and combine their votes)
- Balanced class_weight (Penalises the model for getting M5+ wrong more heavily)

What we found

	precision	recall	f1-score	support
M3-4	0.88	0.97	0.92	12459
M4-5	0.55	0.25	0.34	2118
M5+	0.19	0.03	0.05	225
accuracy			0.85	14802
macro avg	0.54	0.42	0.44	14802
weighted avg	0.82	0.85	0.83	14802

Quick-Simple Definitions:

Precision - How often the ring is right when it goes off

Recall - How often it rings when there is actually an earthquake

Accuracy - How often was it right overall

Macro avg - How well did it treat every class fairly

Weighted avg - How well did it do, weighted by how common each class is

1. M3-4 is performing well. 0.92 F1 is solid.
2. M4-5 and M5+ are struggling badly (0.35 and 0.05 F1)
3. The model is essentially guessing M3-4 most of the time and occasionally getting lucky on M4-5

Let's read the M5+ row specifically:

Precision 0.19 - when it predicts M5+, it's wrong 81% of the time

Recall 0.03 - it only catches 3% of actual M5+ events

What this means

Because our three features — latitude, longitude, depth — simply don't contain enough signal to reliably predict magnitude class. Location and depth constrain what type of earthquake can occur, but they don't precisely determine how big it will be. Magnitude has a large random component that no spatial model can fully capture.

The model demonstrates that spatial and depth features alone are insufficient predictors of magnitude class, suggesting magnitude is not purely a function of location.

Further Improvements/Future Work

Many of these extensions may not be immediately feasible due to computational limits or the need for deeper expertise. They can be considered in the future if access to more powerful hardware becomes available or further study into advanced spatial-temporal methods is undertaken.

Pareto principle (20-80) was used to rank the improvements by their gain and effort.

Highest Gain, Low Effort

1. Improved Machine Learning Pipeline

- The Random Forest model used only raw event coordinates and depth as features, leaving derived spatial outputs from earlier pipeline stages (per-cell trend slopes, local density estimates) disconnected from the predictive model.
- Future work would connect these richer features to the classifier. Additionally, replacing the current chronological train/test split with spatially-blocked cross-validation would produce a more honest evaluation of model generalisation across geographic regions.
- **Regional Model Stratification**
A single Random Forest model was trained across all of New Zealand, treating the entire country as one homogeneous seismic environment. In reality, New Zealand's North and South Islands have fundamentally different tectonic characteristics — the North Island is dominated by the Hikurangi subduction zone while the South Island is primarily governed by the Alpine Fault system. Training separate regional models for each tectonic setting could capture localised seismic patterns more accurately than a single global model, potentially improving classification performance particularly for the underrepresented M5+ class.

2. Risk and Impact Assessment

- Extending the analysis to estimate shaking intensity, population exposure, or infrastructure risk could provide actionable insights for emergency management and urban planning.

Medium Gain, Medium Effort

1. Inclusion of Smaller Magnitude Events

- Incorporating earthquakes below magnitude 3.0 could provide a more complete picture of seismic activity and microseismic clustering

2. Dynamic Grid or Adaptive Resolution

- Using a variable grid size based on seismic density could capture fine-scale patterns in highly active areas while maintaining efficiency elsewhere.

3. Advanced Density Estimation

- Replacing the current histogram-based smoothing with true space-time KDE would more accurately capture both spatial clustering and temporal evolution of seismic activity simultaneously, as opposed to treating space and time as separate analyses.

Low Gain, High Effort

1. Integration with Fault and Geology Data

- Combining seismic data with detailed geological maps, soil types, and fault slip rates could improve interpretation of trends and highlight potential blind faults.
- While this integration would meaningfully strengthen the geological interpretation of spatial trends, it requires domain expertise in geoscience beyond the current scope of this project and is considered a longer-term extension as technical knowledge develops.

2. Advanced Statistical Modelling

- Bayesian hierarchical models or machine learning-based hotspot detection could better account for aftershock sequences and temporal clustering, though these require significantly deeper statistical expertise and are considered longer-term extensions beyond the current project scope.

Acknowledgements

This report acknowledges GeoNet for providing the earthquake data that made this analysis possible and also wishes to thank the computational tools and libraries — including Python, Jupyter Notebook, and various geospatial and statistical packages — which enabled the data processing and analysis. AI tools were used throughout the project to assist with debugging, research, and problem-solving.